OOP & OOAD TRAINING PROJECT

Library Management System

A practical project where participants design and build a simple library system used by a librarian to search books, issue books to members, return borrowed books, and view member/book details.

← Back to main course

1 Primary user: Librarian

5 Main features

6 Suggested screens

OOP Book, Member, BorrowRecord

Overview · Features · Domain Objects · UI Mocks · User Flows · Business Rules · Data Fields · OOP/OOAD Guide

1 Product Overview

Project Purpose

The purpose of this project is to apply the OOP and OOAD concepts participants learned during the training sessions. They will practice identifying classes, responsibilities, relationships, use cases, business rules, and UI flows.

Main User

The system is used directly by the **Librarian**. Members do not log in to the system. A member requests a book or returns a book physically, and the librarian performs the required operation in the system.

Problem Scenario

A member requests the librarian to borrow a book from the library. The librarian searches the book by title or author name. The system displays all matching books with availability status. If a book copy is available, the librarian issues it to the member. When a member returns a book, the librarian can search by ISBN/copy ID or open the member profile and return the selected borrowed book.

2 Main Features

🔎

Search Book

Search books by title, author name, or ISBN and see copy-wise availability.

📘

Issue Book

Issue an available book copy to a valid member and generate a borrow record.

↩

Return Book

Return a borrowed book and make the copy available again.

👤

Member Details

View member information, currently borrowed books, and borrowing history.

📖

Book Details

View book information, all copies, statuses, borrower name, and due date.

🧾

Summary Popup

Show a clear confirmation message after issuing or returning a book.

3 Domain Objects

> **A real library usually has multiple copies of the same book. So, the system should distinguish between Book and BookCopy.**

Book
- BookId
- Title
- Author
- ISBN
- Category

BookCopy
- CopyId
- BookId
- Status
- Available / Borrowed

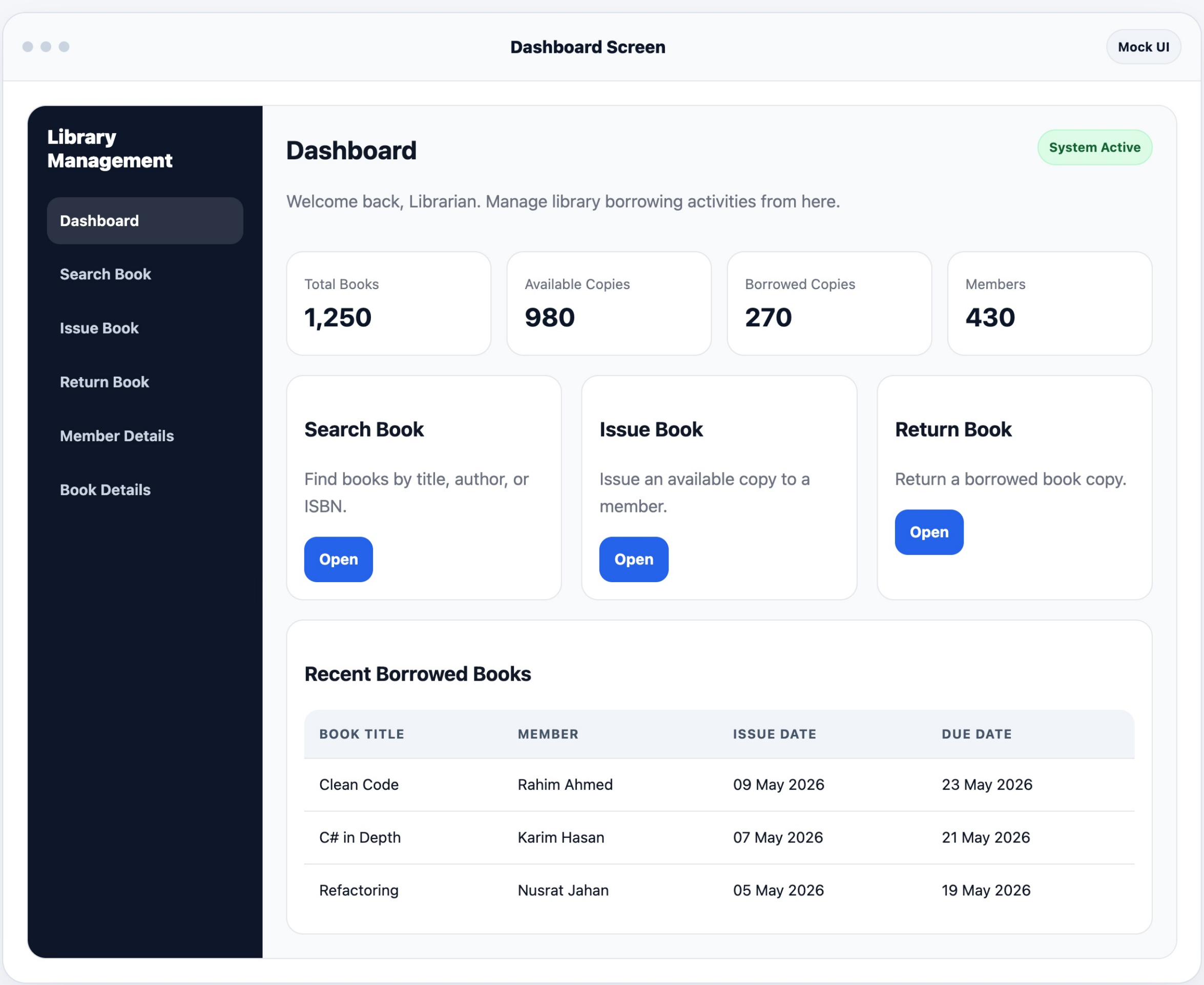
Member
- MemberId
- Name
- Phone
- Address
- Status

BorrowRecord
- BorrowId
- CopyId
- MemberId
- IssueDate
- DueDate
- ReturnDate

Librarian
- Searches book
- Issues book
- Returns book
- Views details

LibraryService
- SearchBook()
- IssueBook()
- ReturnBook()
- GetMemberDetails()

4 UI Mocks



Dashboard Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

Dashboard System Active

Welcome back, Librarian. Manage library borrowing activities from here.

Total Books 1,250 · Available Copies 980 · Borrowed Copies 270 · Members 430

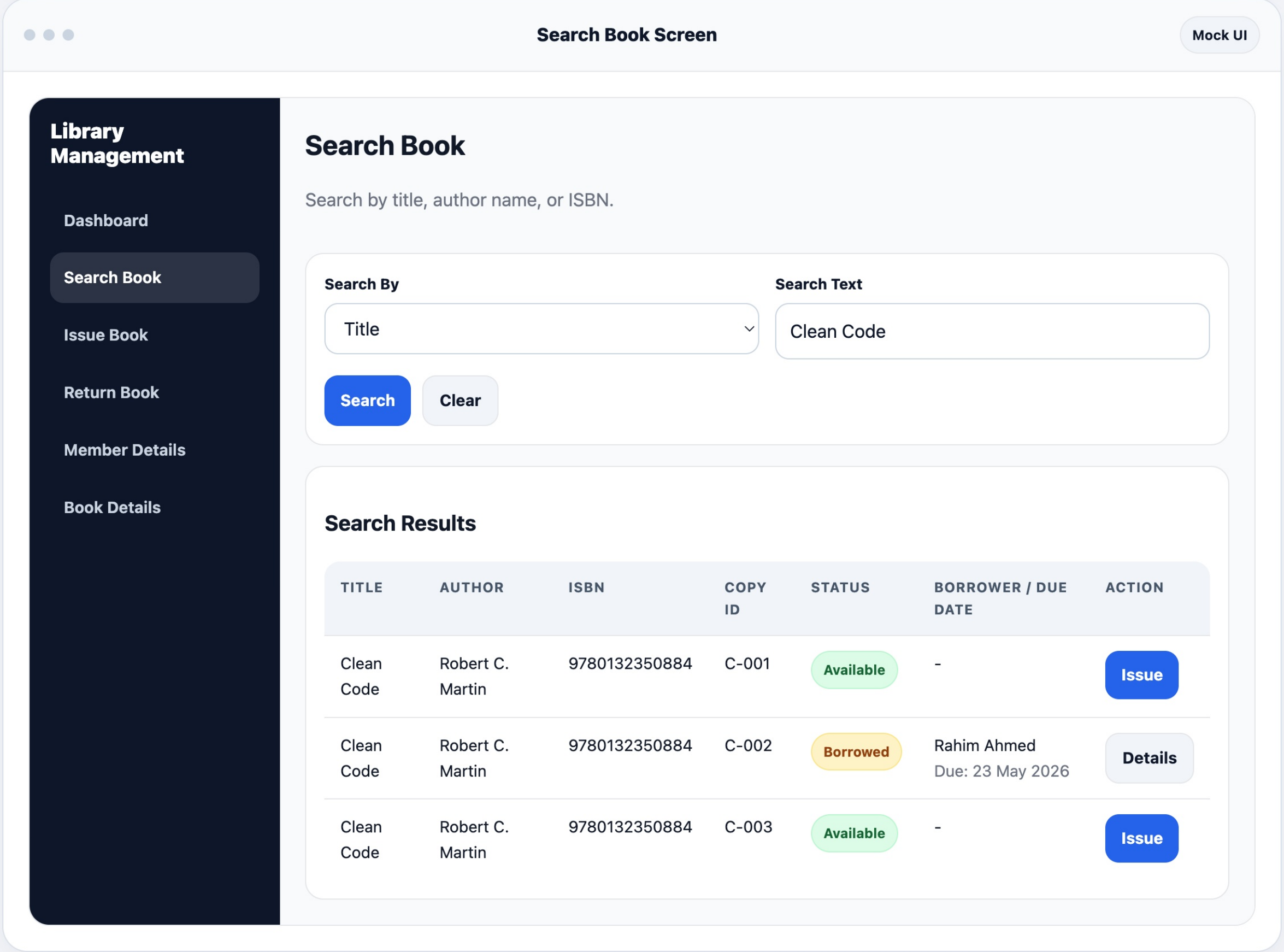
Search Book Find books by title, author, or ISBN. Open

Issue Book Issue an available copy to a member. Open

Return Book Return a borrowed book copy. Open

Recent Borrowed Books

BOOK TITLE	MEMBER	ISSUE DATE	DUE DATE
Clean Code	Rahim Ahmed	09 May 2026	23 May 2026
C# in Depth	Karim Hasan	07 May 2026	21 May 2026
Refactoring	Nusrat Jahan	05 May 2026	19 May 2026



Search Book Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

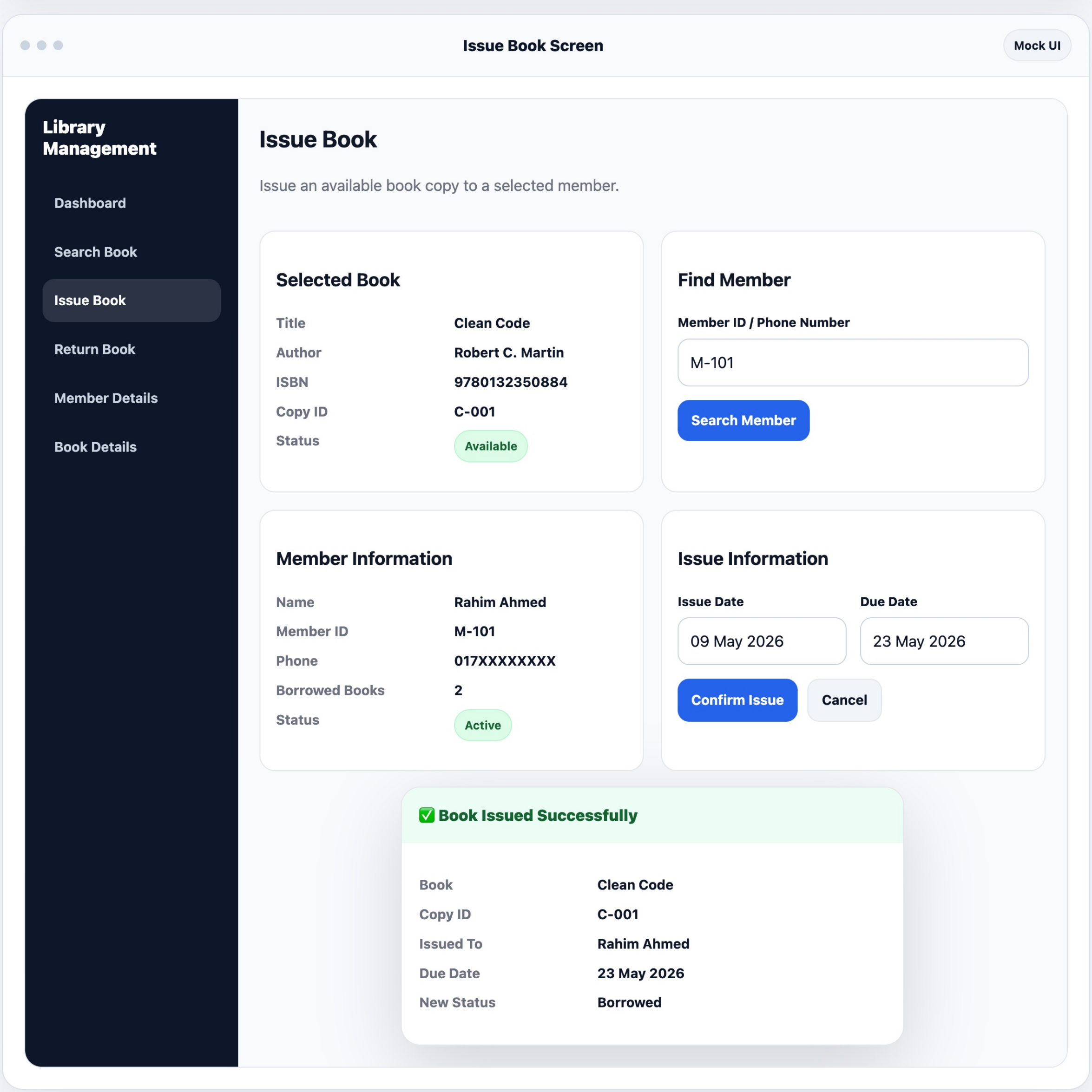
Search Book

Search by title, author name, or ISBN.

Search By: Title · Search Text: Clean Code · Search · Clear

Search Results

TITLE	AUTHOR	ISBN	COPY ID	STATUS	BORROWER / DUE DATE	ACTION
Clean Code	Robert C. Martin	9780132350884	C-001	Available	-	Issue
Clean Code	Robert C. Martin	9780132350884	C-002	Borrowed	Rahim Ahmed Due: 23 May 2026	Details
Clean Code	Robert C. Martin	9780132350884	C-003	Available	-	Issue



Issue Book Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

Issue Book

Issue an available book copy to a selected member.

Selected Book

Title	Clean Code
Author	Robert C. Martin
ISBN	9780132350884
Copy ID	C-001
Status	Available

Find Member

Member ID / Phone Number: M-101 · Search Member

Member Information

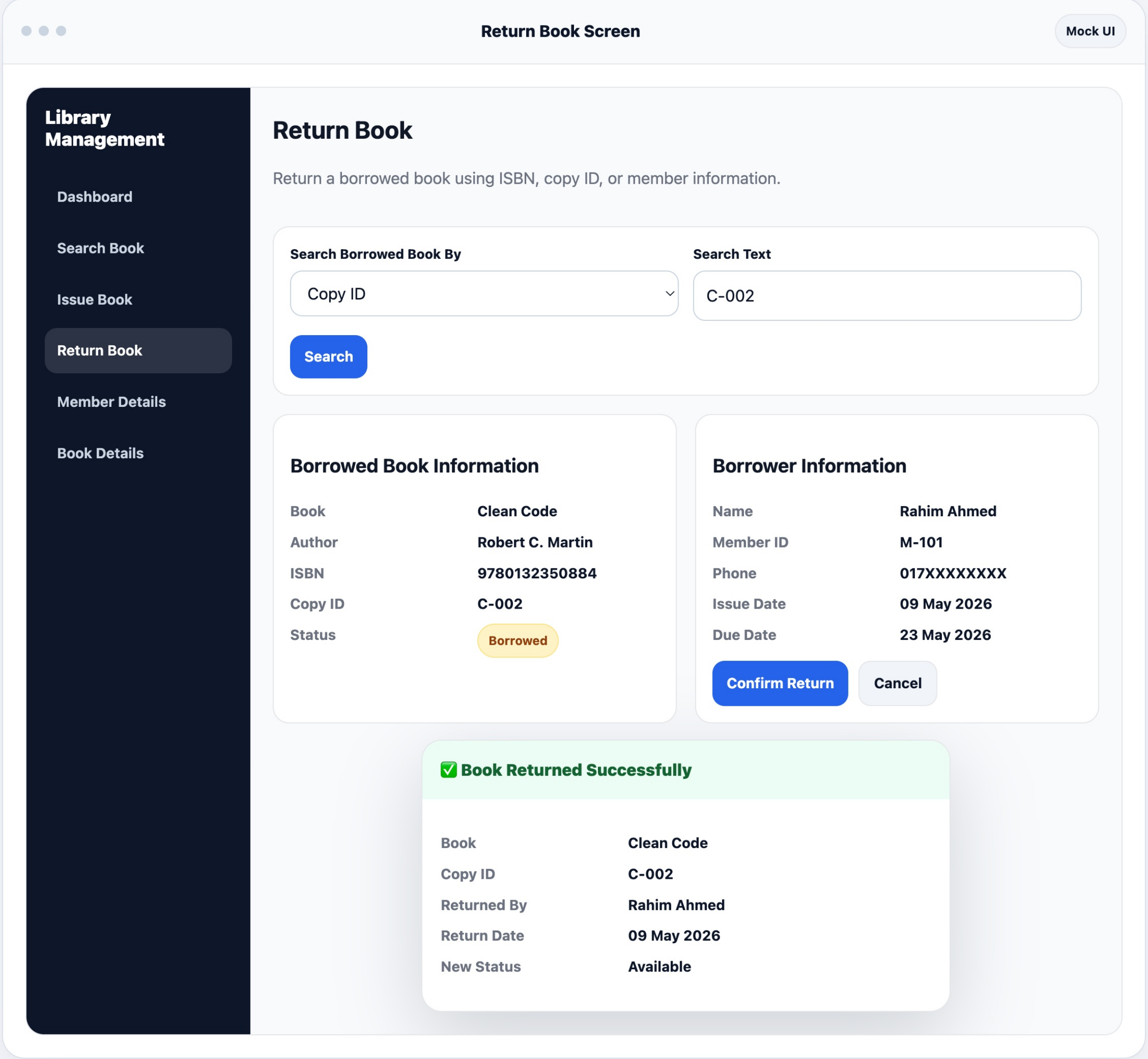
Name	Rahim Ahmed
Member ID	M-101
Phone	017XXXXXXXX
Borrowed Books	2
Status	Active

Issue Information

Issue Date: 09 May 2026 · Due Date: 23 May 2026 · Confirm Issue · Cancel

✅ **Book Issued Successfully**

Book	Clean Code
Copy ID	C-001
Issued To	Rahim Ahmed
Due Date	23 May 2026
New Status	Borrowed



Return Book Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

Return Book

Return a borrowed book using ISBN, copy ID, or member information.

Search Borrowed Book By: Copy ID · Search Text: C-002 · Search

Borrowed Book Information

Book	Clean Code
Author	Robert C. Martin
ISBN	9780132350884
Copy ID	C-002
Status	Borrowed

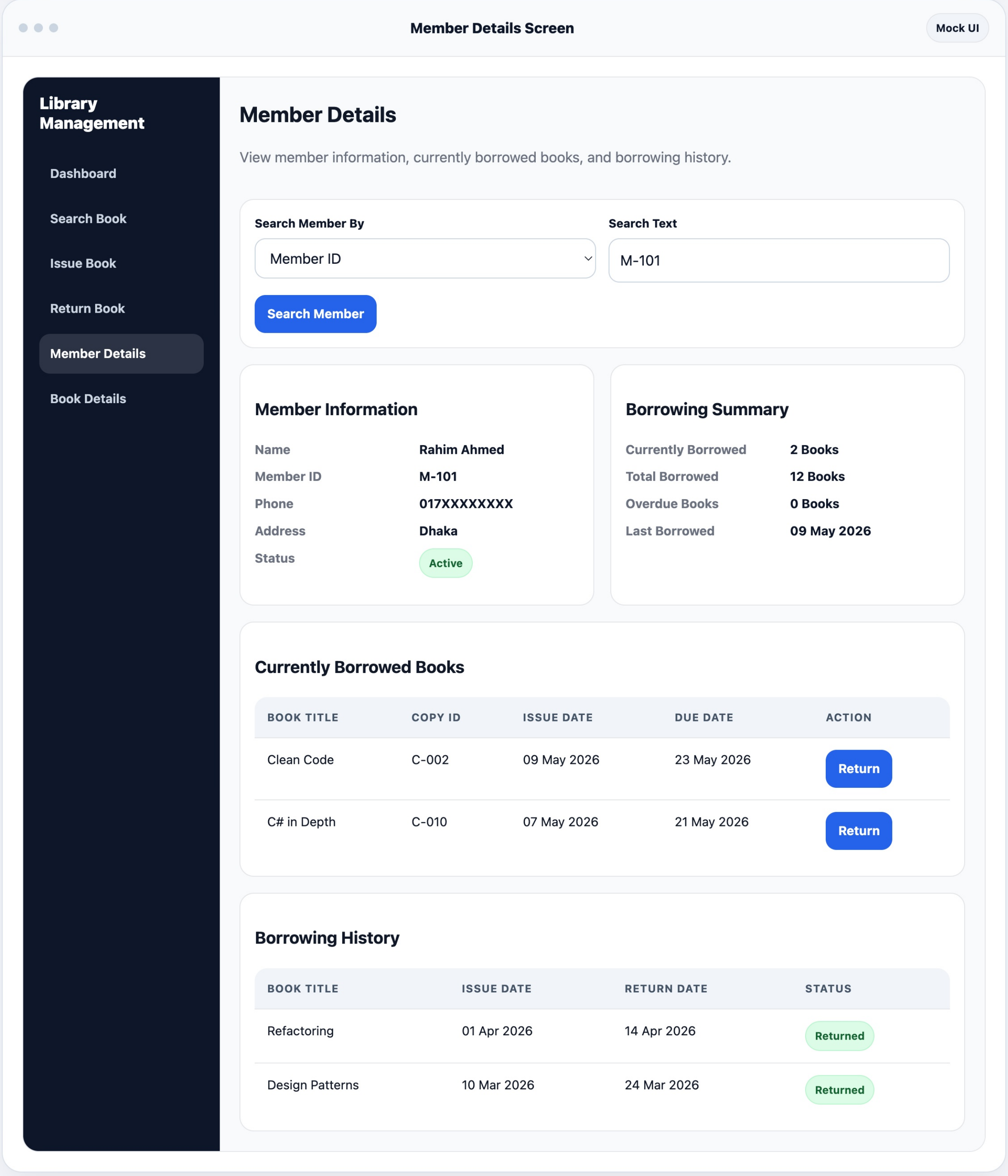
Borrower Information

Name	Rahim Ahmed
Member ID	M-101
Phone	017XXXXXXXX
Issue Date	09 May 2026
Due Date	23 May 2026

Confirm Return · Cancel

✅ **Book Returned Successfully**

Book	Clean Code
Copy ID	C-002
Returned By	Rahim Ahmed
Return Date	09 May 2026
New Status	Available



Member Details Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

Member Details

View member information, currently borrowed books, and borrowing history.

Search Member By: Member ID · Search Text: M-101 · Search Member

Member Information

Name	Rahim Ahmed
Member ID	M-101
Phone	017XXXXXXXX
Address	Dhaka
Status	Active

Borrowing Summary

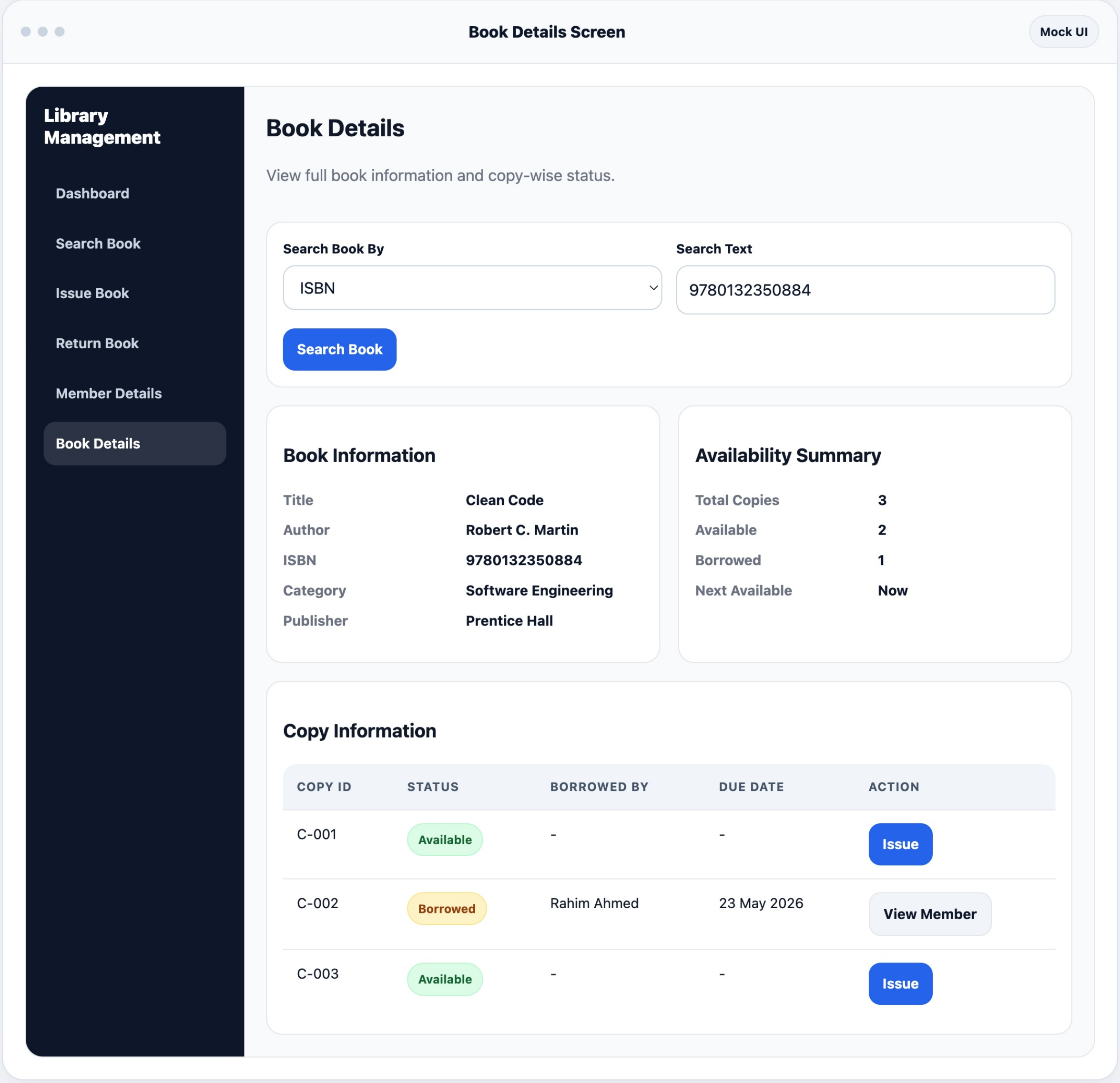
Currently Borrowed	2 Books
Total Borrowed	12 Books
Overdue Books	0 Books
Last Borrowed	09 May 2026

Currently Borrowed Books

BOOK TITLE	COPY ID	ISSUE DATE	DUE DATE	ACTION
Clean Code	C-002	09 May 2026	23 May 2026	Return
C# in Depth	C-010	07 May 2026	21 May 2026	Return

Borrowing History

BOOK TITLE	ISSUE DATE	RETURN DATE	STATUS
Refactoring	01 Apr 2026	14 Apr 2026	Returned
Design Patterns	10 Mar 2026	24 Mar 2026	Returned



Book Details Screen Mock UI

Library Management: Dashboard · Search Book · Issue Book · Return Book · Member Details · Book Details

Book Details

View full book information and copy-wise status.

Search Book By: ISBN · Search Text: 9780132350884 · Search Book

Book Information

Title	Clean Code
Author	Robert C. Martin
ISBN	9780132350884
Category	Software Engineering
Publisher	Prentice Hall

Availability Summary

Total Copies	3
Available	2
Borrowed	1
Next Available	Now

Copy Information

COPY ID	STATUS	BORROWED BY	DUE DATE	ACTION
C-001	Available	-	-	Issue
C-002	Borrowed	Rahim Ahmed	23 May 2026	View Member
C-003	Available	-	-	Issue

5 User Flows

Overall Flow

Dashboard → Search Book → Check Availability → Issue Book → Confirmation

Flow 1: Search Book

1. Librarian opens the Search Book screen.
2. Librarian selects search type: title, author, or ISBN.
3. Librarian enters search text.
4. System displays matching books.
5. System shows copy-wise status.
6. Librarian can issue an available copy or view details.

Flow 2: Issue Book

1. Librarian selects an available book copy.
2. System opens the Issue Book screen.
3. Librarian searches member by ID or phone.
4. System displays member information.
5. Librarian confirms due date.
6. Librarian clicks Confirm Issue.
7. System creates a borrow record.
8. Book copy status becomes Borrowed.

Flow 3: Return Book by Copy ID / ISBN

1. Librarian opens Return Book screen.
2. Librarian enters copy ID or ISBN.
3. System displays borrowed book information.
4. System displays borrower details.
5. Librarian clicks Confirm Return.
6. System updates the borrow record.
7. Book copy status becomes Available.

Flow 4: Return Book from Member Profile

1. Librarian opens Member Details screen.
2. Librarian searches member.
3. System displays currently borrowed books.
4. Librarian clicks Return beside a book.
5. System displays selected book details.
6. Librarian confirms return.
7. System removes the book from current borrowed list.

Flow 5: View Member Details

1. Librarian searches member by ID, phone, or name.
2. System displays member profile.
3. System displays currently borrowed books.
4. System displays borrowing history.
5. Librarian can return a borrowed book from this screen.

Flow 6: View Book Details

1. Librarian searches book by ISBN, title, or author.
2. System displays book information.
3. System displays all physical copies.
4. System shows available and borrowed copies.
5. Librarian can issue available copy or view borrower information.

6 Business Rules

Issue Book Rules

- A book copy can be issued only if it is available.
- A borrowed copy cannot be issued again.
- A valid active member is required to issue a book.
- Due date must be greater than issue date.
- After issuing, the copy status becomes Borrowed.
- A borrow record must be created.

Return Book Rules

- Only borrowed books can be returned.
- Return date must be recorded.
- After return, the copy status becomes Available.
- The book should be removed from the member's current borrowed list.
- The borrowing history should still keep the old record.

Search Rules

- Librarian can search by title, author, or ISBN.
- Search result should show all matching books.
- Search result should show availability status.
- If borrowed, result should show borrower name and due date.

Member Rules

- Only active members can borrow books.
- Member profile should show currently borrowed books.
- Member profile should show borrowing history.
- Librarian can return a book from the member profile.

7 Minimum Data Fields

Book

- BookId
- Title
- Author
- ISBN
- Category
- Publisher

BookCopy

- CopyId
- BookId
- Status
- Available
- Borrowed

Member

- MemberId
- Name
- Phone
- Address
- Status

BorrowRecord

- BorrowId
- CopyId
- MemberId
- IssueDate
- DueDate
- ReturnDate
- Status

8 OOP / OOAD Practice Guide

Suggested Beginner Classes

- Book
- Member
- BorrowRecord
- Library

This version is easier for beginners. They can start by storing books and members in lists.

Suggested Improved Classes

- Book
- BookCopy
- Member
- BorrowRecord
- BookService
- MemberService
- BorrowService

This version is more practical because it supports multiple copies of the same book.

Participant Tasks

1. Identify all classes from the problem statement.
2. Write responsibilities for each class.
3. Draw relationships among Book, BookCopy, Member, and BorrowRecord.
4. Write use cases for Search Book, Issue Book, Return Book, Member Details, and Book Details.
5. Build the UI screens using simple HTML/CSS or any framework.
6. Implement the core logic using OOP principles.
7. Test the important business rules.

> **Recommended project scope: keep the first version simple. Use in-memory collections or simple database tables. Focus on object design, responsibilities, flows, and business rules before adding complex features.**

See the main program